

Complete Publication List

Oliver Gutsche

July 01, 2026

1. Search for long-lived particles decaying into muons in proton-proton collisions at $\sqrt{s} = 13.6$ TeV using data scouting

Gevorgyan, Arzunik et al.

[arXiv:2606.25840](#) | 2026-06-24

2. Observation of electroweak production of pairs of Z bosons in proton-proton collisions at 13 TeV

Gevorgyan, Arzunik et al.

[arXiv:2606.20558](#) | 2026-06-18

3. Analysis of the CP structure of the Yukawa coupling between the Higgs boson and tau leptons in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2606.03510](#) | 2026-06-02

4. Search for single production of a vector-like B' quark decaying to a top quark and a W boson in the single-lepton final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tauqeer, Komal et al.

[arXiv:2606.01423](#) | 2026-05-31

5. Dependence of two-particle azimuthal correlations on the forward rapidity gap width in pPb collisions at $\sqrt{s_{NN}} = 8.16$ TeV

Hayrapetyan, Aram et al.

[arXiv:2606.00171](#) | 2026-05-29

6. Search for light scalar particles produced in Higgs boson decays in exclusive final states with two muons and two hadrons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2605.27875](#) | 2026-05-27

7. Search for a new heavy scalar resonance decaying to a pair of Z bosons in the four-lepton final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2605.26462](#) | 2026-05-26

8. Search for low-mass resonances decaying to $\tau\tau$ and measurement of the $Y \rightarrow \tau\tau$ decay in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.

[arXiv:2605.25103](#) | 2026-05-24

9. Studies of $Z \rightarrow 4\ell$ decays in proton-proton collisions at $\sqrt{s} = 8$ and 13 TeV

Tumasyan, Armen et al.

[arXiv:2605.25158](#) | 2026-05-24

10. Search for Higgs boson decays into two neutral scalars with unequal masses in final states with b quarks and tau leptons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2605.19144](#) | 2026-05-18

11. Search for the nonresonant production of a pair of additional Higgs bosons in the Type-X two-Higgs-doublet model in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.
[arXiv:2605.16655](#) | 2026-05-15

12. Search for new physics in triple boson production in proton-proton collisions at $\sqrt{s} = 13$ TeV using the effective field theory approach

Hayrapetyan, Aram et al.
[arXiv:2605.15023](#) | 2026-05-14

13. Constraints on anomalous Higgs boson couplings to vector bosons and fermions using the $\gamma\gamma$ final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.
[arXiv:2605.14245](#) | 2026-05-14

14. First measurements of vector boson scattering in $W^{\pm}W^{\pm}$ and WZ production in all-leptonic final states at $\sqrt{s} = 13.6$ TeV

Gevorgyan, Arzunik et al.
[arXiv:2605.15396](#) | 2026-05-14

15. Search for pair production of additional neutral scalars within the Inert Doublet Model in a final state with two electrons or two muons in proton-proton collisions at $\sqrt{s} = 13$ TeV and 13.6 TeV

Gevorgyan, Arzunik et al.
[arXiv:2605.13614](#) | 2026-05-13

16. Search for single vector-like quark production in opposite-sign dilepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.
[arXiv:2605.13649](#) | 2026-05-13

17. Search for a new heavy scalar resonance decaying into the Higgs boson and a new scalar particle in the $b\bar{b}b\bar{b}$ final state using proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.
[arXiv:2605.02848](#) | 2026-05-04

18. Observation of the rare decay $\eta \rightarrow \mu^+ \mu^- e^+ e^-$

Hayrapetyan, Aram et al.
[arXiv:2605.00615](#) | 2026-05-01

19. Search for light charged Higgs bosons decaying to charm and strange quarks in $t\bar{t}$ events in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.
[arXiv:2604.27795](#) | 2026-05-01

20. Improved results on Higgs boson pair production in the $4b$ final state

Hayrapetyan, Aram et al.
[arXiv:2604.27044](#) | 2026-04-29

21. Search for electroweakinos in compressed-spectrum scenarios with low-momentum isolated tracks in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.
[arXiv:2604.25604](#) | 2026-04-28

22. Simultaneous measurements of N-subjettiness observables in jets from gluons and light-flavour quarks, and in decays of boosted W bosons and top quarks

Hayrapetyan, Aram et al.
[arXiv:2604.25538](#) | 2026-04-28

23. Measurement of the $Z \rightarrow \mu^+ \mu^-$ angular coefficients in pp collisions at $\sqrt{s} = 13$ TeV as functions of transverse momentum and rapidity

Gevorgyan, Arzunik et al.

[arXiv:2604.25678](#) | 2026-04-28

24. Search for associated production of a Higgs boson and two vector bosons via vector boson scattering at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.24531](#) | 2026-04-27

25. Search for dark matter produced in association with a dark Higgs boson decaying into a bottom quark-antiquark pair in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.20999](#) | 2026-04-22

26. Search for the single production of vector-like quarks decaying into a W boson and a b quark using single-lepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.17564](#) | 2026-04-19

27. Highly boosted dielectron identification in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.13320](#) | 2026-04-14

28. Search for heavy resonances decaying into four-lepton final states via light bosons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.14236](#) | 2026-04-14

29. A search for microscopic black holes, string balls, and sphalerons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.10732](#) | 2026-04-12

30. Search for a new heavy resonance decaying to a top quark and a neutral scalar boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.10729](#) | 2026-04-12

31. Particle transformers for identifying Lorentz-boosted Higgs bosons decaying to a pair of W bosons

Hayrapetyan, Aram et al.

[arXiv:2604.09809](#) | 2026-04-10

32. Observation of nuclear suppression in coherent $\Upsilon(1S)$ photoproduction off heavy nuclei at the LHC

Belyaev, Andrey et al.

[arXiv:2604.05814](#) | 2026-04-07

33. Search for soft unclustered energy patterns produced in association with a W or Z boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.05996](#) | 2026-04-07

34. Evidence of $ZZ\gamma$ production and observation of $4\ell\gamma$ in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

35. Search for Higgs boson pair production in the $b\bar{b}WW$ decay channel with two leptons in the final state using proton-proton collision data at $\sqrt{s} = 13.6$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2604.02127](#) | 2026-04-02

36. Measurement of dijet angular distributions and search for beyond the standard model physics in proton-proton collisions at $\sqrt{s} = 13$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2603.25458](#) | 2026-03-26

37. Search for new particles decaying into top quark-antiquark pairs in proton-proton collisions at $\sqrt{s} = 13$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2603.23454](#) | 2026-03-24

38. Search for low-mass vector and scalar resonances decaying into a quark-antiquark pair in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2603.21965](#) | 2026-03-23

39. Search for Higgs boson production at high transverse momentum in the WW decay channel in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2603.22233](#) | 2026-03-23

40. Measurement of the jet mass in hadronic decays of boosted W bosons at 13 TeV and extraction of the W boson mass

Hayrapetyan, Aram et al.

[arXiv:2603.19963](#) | 2026-03-20

41. Jet peak shapes based on two-particle angular correlations in lead-lead collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Belyaev, Andrey et al.

[arXiv:2603.14385](#) | 2026-03-15

42. Measurement of the t -channel single top quark cross section in proton-proton collisions at $\sqrt{s} = 5.02$ TeV

Hayrapetyan, Aram et al.

[arXiv:2603.13592](#) | 2026-03-13

43. Search for Z' bosons decaying into charginos in final states with two oppositely charged leptons and missing transverse momentum in pp collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2603.11035](#) | 2026-03-11

44. Measurement of angular correlations inside jets induced by gluon polarization in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.

[arXiv:2603.03689](#) | 2026-03-04

45. Search for a narrow resonance with a mass between 10 and 70 GeV decaying to a pair of photons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2603.03250](#) | 2026-03-03

46. Combination of ATLAS and CMS searches for Higgs boson pair production at $\sqrt{s} = 13$ TeV

Aad, Georges et al.

[arXiv:2602.23991](#) | 2026-02-27

47. Search for pair production of heavy resonances in final states with a photon and large-radius jets in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

Phys.Rev.D 113 (2026) 112005 | [arXiv:2602.20477](#) | DOI:10.1103/bwqs-vgqn | 2026-02-24

48. System-size dependence of charged-particle suppression in ultrarelativistic nucleus-nucleus collisions

Belyaev, Andrey et al.

[arXiv:2602.21325](#) | 2026-02-24

49. Observation of the jet diffusion wake using dijets in heavy ion collisions

Hayrapetyan, Aram et al.

[arXiv:2602.19431](#) | 2026-02-23

50. First measurement of jet axis decorrelation with photon-tagged jets in pp and PbPb collisions at 5.02 TeV

Hayrapetyan, Aram et al.

[arXiv:2602.18279](#) | 2026-02-20

51. Search for a new resonance decaying to a Higgs boson and a scalar boson in events with two b jets and two Z bosons in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.

[arXiv:2602.18223](#) | 2026-02-20

52. Combined measurements and interpretations of Higgs boson production and decay in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2602.18611](#) | 2026-02-20

53. Measurement of event shape variables using charged particles inside jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2602.17509](#) | 2026-02-19

54. Building an AI-native Research Ecosystem for Experimental Particle Physics: A Community Vision

Aarrestad, Thea Klæboe et al.

[arXiv:2602.17582](#) | 2026-02-19

55. Interpretable Models for Workflow Differentiation in High-Performance Scientific Networks

Giannakou, Anna et al.

(2026) 349 | DOI:10.1109/ICNC68183.2026.11416957 | 2026

56. Influence of strangeness on the anisotropic flow of prompt $D^{\pm}_s$ mesons in PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Belyaev, Andrey et al.

[arXiv:2602.14221](#) | 2026-02-15

57. High-level hadronic tau lepton triggers of the CMS experiment in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

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58. Measurement of B meson production fraction ratios in proton-proton collisions at $\sqrt{s} = 13$ TeV using open-charm and charmonium decays

Hayrapetyan, Aram et al.

[arXiv:2602.10270](#) | 2026-02-10

59. Probing early parton emissions in heavy ion collisions using the Lund jet plane

Belyaev, Andrey et al.

[arXiv:2602.09271](#) | 2026-02-09

60. Observation of a family of all-charm tetraquarks

Hayrapetyan, Aram et al.

[arXiv:2602.02252](#) | 2026-02-02

61. Observation of $\Upsilon(1S) + Z$ associated production and measurement of the effective double-parton scattering cross section in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2602.02770](#) | 2026-02-02

62. Search for heavy scalar resonances decaying to Lorentz-boosted Higgs and Higgs-like bosons in the $b\bar{b}4q$ final state at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2602.00273](#) | 2026-01-30

63. Measurement and effective field theory interpretation of the photon-fusion production cross section of a pair of W bosons in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[JHEP 06 \(2026\) 187](#) | [arXiv:2601.21574](#) | [DOI:10.1007/JHEP06\(2026\)187](#) | 2026-01-29

64. Search for heavy long-lived charged particles with level-1 trigger scouting data from proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.

[Phys.Lett.B 878 \(2026\) 140498](#) | [arXiv:2601.20063](#) | [DOI:10.1016/j.physletb.2026.140498](#) | 2026-01-27

65. Measurement of the $\Upsilon(1S)$, $\Upsilon(2S)$, and $\Upsilon(3S)$ differential cross sections in pp collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.20023](#) | 2026-01-27

66. Search for heavy resonances decaying into two Higgs bosons in the $b\bar{b}\tau^+\tau^-$ final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.20011](#) | 2026-01-27

67. Evaluating Application Characteristics for GPU Portability Layer Selection

Atif, Mohammad et al.

[arXiv:2601.17526](#) | 2026-01-24

68. Search for the pair production of long-lived supersymmetric partners of the tau lepton in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.17576](#) | 2026-01-24

69. Full event interpretation with machine-learning-based particle-flow reconstruction in the CMS detector

Hayrapetyan, Aram et al.

[arXiv:2601.17554](#) | 2026-01-24

70. Strategy and performance of the CMS long-lived particle trigger program in proton-proton collisions at $\sqrt{s} = 13.6$ TeV
Hayrapetyan, Aram et al.
[arXiv:2601.17544](#) | 2026-01-24
71. Measurement of the $Z\gamma$ production cross section and search for anomalous neutral triple gauge couplings in pp collisions at $\sqrt{s} = 13$ TeV
Hayrapetyan, Aram et al.
[arXiv:2601.14102](#) | 2026-01-20
72. Combination of searches for heavy vector boson resonances in proton-proton collisions at $\sqrt{s} = 13$ TeV
Hayrapetyan, Aram et al.
[arXiv:2601.12583](#) | 2026-01-18
73. Search for dark matter produced in association with a Higgs boson decaying to bottom quarks in proton-proton collisions at $\sqrt{s} = 13$ TeV
Hayrapetyan, Aram et al.
[Phys.Rev.D 113 \(2026\) 092016](#) | [arXiv:2601.11330](#) | [DOI:10.1103/2m87-3dlr](#) | 2026-01-16
74. Measurement of the Higgs boson total decay width using the $H \rightarrow WW \rightarrow e\nu\mu\nu$ decay channel in proton-proton collisions at $\sqrt{s}=13\text{TeV}$
Hayrapetyan, Aram et al.
[Phys.Rev.D 113 \(2026\) 092014](#) | [arXiv:2601.05168](#) | [DOI:10.1103/rny9-zby1](#) | 2026-01-08
75. Search for a boosted Higgs boson decaying to bottom quark pairs in association with a W or Z boson in proton-proton collisions at $\sqrt{s} = 13$ TeV
Hayrapetyan, Aram et al.
[Phys.Lett.B 879 \(2026\) 140534](#) | [arXiv:2601.05362](#) | [DOI:10.1016/j.physletb.2026.140534](#) | 2026-01-08
76. Search for exotic Higgs boson decays $H \rightarrow \mathcal{A}\mathcal{A}$ with $\mathcal{A}\mathcal{A} \rightarrow \gamma\gamma$ in events with a semi-merged topology in proton-proton collisions at $\sqrt{s} = 13$ TeV
Hayrapetyan, Aram et al.
[arXiv:2601.00183](#) | 2026-01-01
77. Search for charged Higgs bosons decaying into top and bottom quarks in lepton+jets final states in proton-proton collisions at $\sqrt{s} = 13$ TeV
Hayrapetyan, Aram et al.
[arXiv:2512.24471](#) | 2025-12-30
78. Machine-learning techniques for model-independent searches in dijet final states
Hayrapetyan, Aram et al.
[arXiv:2512.20395](#) | [DOI:10.5281/zenodo.16656501](#) | 2025-12-23
79. Characterization of the quantum state of top quark pairs produced in proton-proton collisions at $\sqrt{s}=13\text{TeV}$ using the beam and helicity bases
Hayrapetyan, Aram et al.
[Phys.Rev.D 113 \(2026\) L051101](#) | [arXiv:2512.17557](#) | [DOI:10.1103/hkgw-dztk](#) | 2025-12-19
80. Measurement of $Z\gamma$ production in proton-proton collisions at $\sqrt{s} = 13.6$ TeV and constraints on neutral triple gauge couplings
Hayrapetyan, Aram et al.
[arXiv:2512.08582](#) | 2025-12-09
81. Measurements of electroweak production of a photon in association with two jets in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[JHEP 05 \(2026\) 020](#) | [arXiv:2512.00502](#) | [DOI:10.1007/JHEP05\(2026\)020](#) | 2025-11-29

82. Search for Light Pseudoscalar Bosons, Pair-Produced in Higgs Boson Decays in the Four-Electron Final State in Proton-Proton Collisions at $\sqrt{s}=13\text{TeV}$

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[Phys.Rev.Lett. 136 \(2026\) 181807](#) | [arXiv:2511.19563](#) | [DOI:10.1103/r53x-8fqp](#) | 2025-11-24

83. Search for new physics in the final state with a single photon and large missing transverse momentum in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 113 \(2026\) 072020](#) | [arXiv:2511.17310](#) | [DOI:10.1103/f9r8-styj](#) | 2025-11-21

84. Search for Higgsinos in final states with low-momentum lepton-track pairs at 13 TeV

Hayrapetyan, Aram et al.

[Phys.Rev.D 113 \(2026\) 092009](#) | [arXiv:2511.16394](#) | [DOI:10.1103/w3b5-92ls](#) | 2025-11-20

85. Search for heavy $H\gamma$ and $Z\gamma$ resonances with a bottom quark-antiquark pair in the final state in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

Hayrapetyan, Aram et al.

[arXiv:2511.14583](#) | 2025-11-18

86. Search for low-mass hidden-valley dark showers with non-prompt muon pairs in proton-proton collisions at $\sqrt{s}=13 \text{ TeV}$

Hayrapetyan, Aram et al.

[JHEP 03 \(2026\) 189](#) | [arXiv:2511.11888](#) | [DOI:10.1007/JHEP03\(2026\)189](#) | 2025-11-14

87. Search for a new neutral gauge boson produced in association with one or two b jets and decaying into a pair of muons in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

Chekhovsky, Vladimir et al.

[arXiv:2511.11853](#) | 2025-11-14

88. Search for long-lived particles using displaced vertices with low-momentum tracks in proton-proton collisions at $\sqrt{s}=13 \text{ TeV}$

Hayrapetyan, Aram et al.

[JHEP 05 \(2026\) 191](#) | [arXiv:2511.08212](#) | [DOI:10.1007/JHEP05\(2026\)191](#) | 2025-11-11

89. Identification of tau leptons using a convolutional neural network with domain adaptation

Hayrapetyan, Aram et al.

[JINST 20 \(2025\) P12032](#) | [arXiv:2511.05468](#) | [DOI:10.1088/1748-0221/20/12/P12032](#) | 2025-11-07

90. Jet fragmentation function and groomed substructure of bottom quark jets in proton-proton collisions at 5.02 TeV

Chekhovsky, Vladimir et al.

[JHEP 04 \(2026\) 147](#) | [arXiv:2511.10666](#) | [DOI:10.1007/JHEP04\(2026\)147](#) | 2025-11-06

91. Inclusive and differential measurements of the $t\bar{t}\gamma$ cross section and the $t\bar{t}\gamma / t\bar{t}$ cross section ratio in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

Hayrapetyan, Aram et al.

[JHEP 04 \(2026\) 086](#) | [arXiv:2511.01995](#) | [DOI:10.1007/JHEP04\(2026\)086](#) | 2025-11-03

92. Characterizing the initial state and dynamical evolution in XeXe and PbPb collisions using multiparticle cumulants

Hayrapetyan, Aram et al.

[Phys.Lett.B 876 \(2026\) 140359](#) | [arXiv:2510.26766](#) | [DOI:10.1016/j.physletb.2026.140359](#) | 2025-10-30

93. Search for single production of a vector-like T quark decaying to a top quark and a neutral scalar boson in the lepton+jets final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2510.25874](#) | 2025-10-29

94. Measurement of the ratio of the $B_c^+ \rightarrow J/\psi \tau^+ \nu_\tau$ and $B_c^+ \rightarrow J/\psi \mu^+ \nu_\mu$ branching fractions using three-prong τ lepton decays

Chekhovsky, Vladimir et al.

[Phys.Rev.D 113 \(2026\) L111101](#) | [arXiv:2510.21559](#) | [DOI:10.1103/mw5c-5rw3](#) | 2025-10-24

95. Search for dijet resonances with data scouting in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[JHEP 04 \(2026\) 133](#) | [arXiv:2510.21641](#) | [DOI:10.1007/JHEP04\(2026\)133](#) | 2025-10-24

96. Observation of tWZ Production at the CMS Experiment

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 136 \(2026\) 081802](#) | [arXiv:2510.19080](#) | [DOI:10.1103/rk6w-1pcl](#) | 2025-10-21

97. Search for dark matter production in association with bottom quarks and a lepton pair in proton-proton collisions at $\sqrt{s}=13$ TeV

Chekhovsky, Vladimir et al.

[JHEP 06 \(2026\) 014](#) | [arXiv:2510.12396](#) | [DOI:10.1007/JHEP06\(2026\)014](#) | 2025-10-14

98. Performance of heavy-flavour jet identification in Lorentz-boosted topologies in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JINST 20 \(2025\) P11006](#) | [arXiv:2510.10228](#) | [DOI:10.1088/1748-0221/20/11/P11006](#) | 2025-10-11

99. Observation of Suppressed Charged-Particle Production in Ultrarelativistic Oxygen-Oxygen Collisions

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 136 \(2026\) 162301](#) | [arXiv:2510.09864](#) | [DOI:10.1103/89sf-9t1x](#) | 2025-10-10

100. Combination of searches for nonresonant Higgs boson pair production in proton-proton collisions at $\sqrt{s}= 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2510.07527](#) | 2025-10-08

101. Wasserstein normalized autoencoder for anomaly detection

Hayrapetyan, Aram et al.

[Mach.Learn.Sci.Tech. 7 \(2026\) 035030](#) | [arXiv:2510.02168](#) | [DOI:10.1088/2632-2153/ae6168](#) | 2025-10-02

102. Observation of long-range collective flow in OO and NeNe collisions and implications for nuclear structure studies

Hayrapetyan, Aram et al.

[arXiv:2510.02580](#) | 2025-10-02

103. Vector boson scattering and anomalous quartic couplings in final states with $\ell \nu q q$ or $\ell \ell q q$ plus jets using proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[JHEP 03 \(2026\) 022](#) | [arXiv:2510.00118](#) | [DOI:10.1007/JHEP03\(2026\)022](#) | 2025-09-30

104. Measurement of the dineutrino system kinematic variables in dileptonic top quark pair production in proton-proton collisions at $\sqrt{s}=13$ TeV

Chekhovsky, Vladimir et al.

105. Simultaneous Probe of the Charm and Bottom Quark Yukawa Couplings Using $t\bar{t}H$ Events

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 136 \(2026\) 011801](#) | [arXiv:2509.22535](#) | [DOI:10.1103/9nwb-splk](#) | 2025-09-26

106. Search for resonances decaying to an anomalous jet and a Higgs boson in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Eur.Phys.J.C 86 \(2026\) 137](#) | [arXiv:2509.13635](#) | [DOI:10.1140/epjc/s10052-025-15176-9](#) | 2025-09-17

107. Measurements of $t\bar{t}W$ differential cross sections and the leptonic charge asymmetry at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[JHEP 03 \(2026\) 083](#) | [arXiv:2509.13512](#) | [DOI:10.1007/JHEP03\(2026\)083](#) | 2025-09-16

108. Improving missing transverse momentum estimation with a deep neural network

Hayrapetyan, Aram et al.

[Phys.Rev.D 113 \(2026\) 072010](#) | [arXiv:2509.12012](#) | [DOI:10.1103/c4z7-tqvc](#) | 2025-09-15

109. Measurement of D^0 Meson Photoproduction in Ultraperipheral Heavy Ion Collisions

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162. Search for a heavy pseudoscalar Higgs boson decaying to a 125 GeV Higgs boson and a Z boson in final states with two tau and two light leptons in proton-proton collisions at $\sqrt{s}=13\text{ TeV}$
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180. Measurements of inclusive and differential cross sections for top quark production in association with a Z boson in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

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181. Measurement of the inclusive $t\bar{t}$ cross section in final states with at least one lepton and additional jets with 302 pb^{-1} of pp collisions at $\sqrt{s} = 5.02 \text{ TeV}$

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182. Search for excited tau leptons in the $\tau\tau\gamma$ final state in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

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183. Search for pair production of heavy particles decaying to a top quark and a gluon in the lepton+jets final state in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

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[Eur.Phys.J.C 85 \(2025\) 342](#) | [arXiv:2410.20601](#) | [DOI:10.1140/epjc/s10052-024-13729-y](#) | 2024-10-27

184. Search for the Higgs boson decays to a ρ^0 , ϕ , or K^0_S meson and a photon in proton-proton collisions at $\sqrt{s}=13 \text{ TeV}$

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Hayrapetyan, A. et al.

187. Search for heavy long-lived charged particles with large ionization energy loss in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[Phys.Rev.D 110 \(2024\) 112016](#) | [arXiv:2409.11067](#) | [DOI:10.1103/PhysRevD.110.112016](#) | 2024-09-17

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Hayrapetyan, Aram et al.

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[Phys.Rev.D 109 \(2024\) L111101](#) | [arXiv:2403.11352](#) | [DOI:10.1103/PhysRevD.109.L111101](#) | 2024-03-17

243. Search for the decay of the Higgs boson to a pair of light pseudoscalar bosons in the final state with four bottom quarks in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[Phys.Rev.Lett. 133 \(2024\) 191902](#) | [arXiv:2403.05311](#) | [DOI:10.1103/PhysRevLett.133.191902](#) | 2024-03-08

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[Eur.Phys.J.C 84 \(2024\) 779](#) | [arXiv:2403.00657](#) | [DOI:10.1140/epjc/s10052-024-12925-0](#) | 2024-03-01

250. Search for heavy neutral leptons in final states with electrons, muons, and hadronically decaying tau leptons in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[Phys.Rev.Lett. 132 \(2024\) 241802](#) | [arXiv:2402.18461](#) | [DOI:10.1103/PhysRevLett.132.241802](#) | 2024-02-28

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[Phys.Rev.D 110 \(2024\) 012004](#) | [arXiv:2402.18658](#) | [DOI:10.1103/PhysRevD.110.012004](#) | 2024-02-28

253. Observation of the $\Xi_b^- \rightarrow \psi(2S) \Xi^-$ decay and studies of the $\Xi_b(5945)^0$ baryon in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 110 \(2024\) 012002](#) | [arXiv:2402.17738](#) | [DOI:10.1103/PhysRevD.110.012002](#) | 2024-02-27

254. Search for long-lived particles using displaced vertices and missing transverse momentum in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

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[Phys.Rev.D 109 \(2024\) 112005](#) | [arXiv:2402.15804](#) | [DOI:10.1103/PhysRevD.109.112005](#) | 2024-02-24

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[Comput.Softw.Big Sci. 8 \(2024\) 17](#) | [arXiv:2402.15366](#) | [DOI:10.1007/s41781-024-00124-1](#) | 2024-02-23

256. Search for long-lived particles decaying to final states with a pair of muons in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

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[JHEP 05 \(2024\) 047](#) | [arXiv:2402.14491](#) | [DOI:10.1007/JHEP05\(2024\)047](#) | 2024-02-22

257. Measurement of Energy Correlators inside Jets and Determination of the Strong Coupling $\alpha_S(m_Z)$

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258. Search for bottom-type vectorlike quark pair production in dileptonic and fully hadronic final states in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

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[Phys.Rev.D 110 \(2024\) 052004](#) | [arXiv:2402.13808](#) | [DOI:10.1103/PhysRevD.110.052004](#) | 2024-02-21

259. Search for exotic decays of the Higgs boson to a pair of pseudoscalars in the $\mu\mu b\bar{b}$ and $\tau\tau b\bar{b}$ final states

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260. Search for a scalar or pseudoscalar dilepton resonance produced in association with a massive vector boson or top quark-antiquark pair in multilepton events at $\sqrt{s}=13\text{TeV}$

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[Phys.Rev.D 110 \(2024\) 012013](#) | [arXiv:2402.11098](#) | [DOI:10.1103/PhysRevD.110.012013](#) | 2024-02-16

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[Phys.Rev.Lett. 134 \(2025\) 131802](#) | [arXiv:2402.09932](#) | [DOI:10.1103/PhysRevLett.134.131802](#) | 2024-02-15

262. Differential cross section measurements for the production of top quark pairs and of additional jets using dilepton events from pp collisions at $\sqrt{s} = 13 \text{ TeV}$

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[JHEP 02 \(2025\) 064](#) | [arXiv:2402.08486](#) | [DOI:10.1007/JHEP02\(2025\)064](#) | 2024-02-13

263. Combination of Measurements of the Top Quark Mass from Data Collected by the ATLAS and CMS Experiments at $\sqrt{s}=7$ and 8 TeV

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[Phys.Rev.Lett. 132 \(2024\) 261902](#) | [arXiv:2402.08713](#) | [DOI:10.1103/PhysRevLett.132.261902](#) | 2024-02-13

264. Search for pair production of scalar and vector leptoquarks decaying to muons and bottom quarks in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

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[Phys.Rev.D 109 \(2024\) 112003](#) | [arXiv:2402.08668](#) | [DOI:10.1103/PhysRevD.109.112003](#) | 2024-02-09

265. Combined search for electroweak production of winos, binos, higgsinos, and sleptons in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

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[Phys.Rev.D 109 \(2024\) 112001](#) | [arXiv:2402.01888](#) | [DOI:10.1103/PhysRevD.109.112001](#) | 2024-02-02

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[Phys.Rev.D 110 \(2024\) 032007](#) | [arXiv:2402.01898](#) | [DOI:10.1103/PhysRevD.110.032007](#) | 2024-02-02

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[EPJ Web Conf. 295 \(2024\) 11003](#) | [arXiv:2401.14221](#) | [DOI:10.1051/epjconf/202429511003](#) | 2024-01-25

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[Phys.Rev.D 109 \(2024\) 112013](#) | [arXiv:2401.14494](#) | [DOI:10.1103/PhysRevD.109.112013](#) | 2024-01-25

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[JHEP 01 \(2025\) 011](#) | [arXiv:2401.11355](#) | [DOI:10.1007/JHEP01\(2025\)011](#) | 2024-01-20

271. Test of lepton flavor universality in $B^{\pm} \rightarrow K^{\pm} \mu^{+} \mu^{-}$ and $B^{\pm} \rightarrow K^{\pm} e^{+} e^{-}$ decays in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[Phys.Rev.Lett. 133 \(2024\) 142301](#) | [arXiv:2312.17103](#) | [DOI:10.1103/PhysRevLett.133.142301](#) | 2023-12-28

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[Nature Commun. 16 \(2025\) 7990](#) | [arXiv:2312.17092](#) | [DOI:10.1038/s41467-025-56200-6](#) | 2023-12-28

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[Eur.Phys.J.C 85 \(2025\) 72](#) | [arXiv:2312.16669](#) | [DOI:10.1140/epjc/s10052-024-13606-8](#) | 2023-12-27

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[Phys.Lett.B 855 \(2024\) 138815](#) | [arXiv:2312.11668](#) | [DOI:10.1016/j.physletb.2024.138815](#) | 2023-12-18

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Hayrapetyan, Aram et al.

[JHEP 03 \(2024\) 105](#) | [arXiv:2312.07484](#) | [DOI:10.1007/JHEP03\(2024\)105](#) | 2023-12-12

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Hayrapetyan, Aram et al.

[Phys.Rev.D 109 \(2024\) 072004](#) | [arXiv:2312.08229](#) | [DOI:10.1103/PhysRevD.109.072004](#) | 2023-12-11

280. Measurement of simplified template cross sections of the Higgs boson produced in association with W or Z bosons in the $H \rightarrow b\bar{b}$ decay channel in proton-proton collisions at $\sqrt{s}=13$ TeV

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[Phys.Rev.D 109 \(2024\) 092011](#) | [arXiv:2312.07562](#) | [DOI:10.1103/PhysRevD.109.092011](#) | 2023-12-09

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[Phys.Rev.D 111 \(2025\) 012009](#) | [arXiv:2312.03199](#) | [DOI:10.1103/PhysRevD.111.012009](#) | 2023-12-05

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[Phys.Lett.B 853 \(2024\) 138633](#) | [arXiv:2312.02371](#) | [DOI:10.1016/j.physletb.2024.138633](#) | 2023-12-04

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Gutsche, Oliver et al.

[EPJ Web Conf. 295 \(2024\) 04050](#) | [arXiv:2312.00772](#) | [DOI:10.1051/epjconf/202429504050](#) | 2023-12-01

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[EPJ Web Conf. 295 \(2024\) 01003](#) | [arXiv:2311.16321](#) | [DOI:10.1051/epjconf/202429501003](#) | 2023-11-27

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Hayrapetyan, Aram et al.

[Phys.Lett.B 850 \(2024\) 138478](#) | [arXiv:2311.03261](#) | [DOI:10.1016/j.physletb.2024.138478](#) | 2023-11-06

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[Phys.Rev.D 110 \(2024\) 012010](#) | [arXiv:2311.02725](#) | [DOI:10.1103/PhysRevD.110.012010](#) | 2023-11-05

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[Phys.Lett.B 852 \(2024\) 138582](#) | [arXiv:2311.00130](#) | [DOI:10.1016/j.physletb.2024.138582](#) | 2023-10-31

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[JHEP 05 \(2024\) 046](#) | [arXiv:2310.19893](#) | [DOI:10.1007/JHEP05\(2024\)046](#) | 2023-10-30

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[Phys.Rev.Lett. 133 \(2024\) 011801](#) | [arXiv:2310.14023](#) | [DOI:10.1103/PhysRevLett.133.011801](#) | 2023-10-21

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Hayrapetyan, Aram et al.

[JHEP 03 \(2024\) 134](#) | [arXiv:2310.12229](#) | [DOI:10.1007/JHEP03\(2024\)134](#) | 2023-10-18

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[JHEP 06 \(2024\) 187](#) | [arXiv:2310.11231](#) | [DOI:10.1007/JHEP06\(2024\)187](#) | 2023-10-17

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Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 132 \(2024\) 121901](#) | [arXiv:2310.05164](#) | [DOI:10.1103/PhysRevLett.132.121901](#) | 2023-10-08

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[JINST 19 \(2024\) P02031](#) | [arXiv:2310.03844](#) | [DOI:10.1088/1748-0221/19/02/P02031](#) | 2023-10-05

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[Phys.Rev.D 109 \(2024\) 112009](#) | [arXiv:2310.03154](#) | [DOI:10.1103/PhysRevD.109.112009](#) | 2023-10-04

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[Phys.Lett.B 850 \(2024\) 138518](#) | [arXiv:2310.03233](#) | [DOI:10.1016/j.physletb.2024.138518](#) | 2023-10-04

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Tumasyan, Armen et al.

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Hayrapetyan, Aram et al.

[Phys.Rev.D 109 \(2024\) 072007](#) | [arXiv:2309.16823](#) | [DOI:10.1103/PhysRevD.109.072007](#) | 2023-09-28

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[JHEP 12 \(2023\) 070](#) | [arXiv:2309.16003](#) | [DOI:10.1007/JHEP12\(2023\)070](#) | 2023-09-27

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[JHEP 05 \(2024\) 042](#) | [arXiv:2309.14442](#) | [DOI:10.1007/JHEP05\(2024\)042](#) | 2023-09-25

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Hayrapetyan, Aram et al.

[JHEP 01 \(2024\) 101](#) | [arXiv:2309.12408](#) | [DOI:10.1007/JHEP01\(2024\)101](#) | 2023-09-21

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Hayrapetyan, Aram et al.

[JINST 19 \(2024\) P05064](#) | [arXiv:2309.05466](#) | [DOI:10.1088/1748-0221/19/05/P05064](#) | 2023-09-11

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[Phys.Rev.Lett. 132 \(2024\) 021803](#) | [arXiv:2309.03501](#) | [DOI:10.1103/PhysRevLett.132.021803](#) | 2023-09-07

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[Eur.Phys.J.C 84 \(2024\) 26](#) | [arXiv:2309.01008](#) | [DOI:10.1140/epjc/s10052-023-12268-2](#) | 2023-09-02

305. Search for a third-generation leptoquark coupled to a τ lepton and a b quark through single, pair, and nonresonant production in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[JHEP 05 \(2024\) 311](#) | [arXiv:2308.07826](#) | [DOI:10.1007/JHEP05\(2024\)311](#) | 2023-08-15

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Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 132 \(2024\) 061801](#) | [arXiv:2308.06143](#) | [DOI:10.1103/PhysRevLett.132.061801](#) | 2023-08-11

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[Eur.Phys.J.C 84 \(2024\) 27](#) | [arXiv:2308.02285](#) | [DOI:10.1140/epjc/s10052-023-12258-4](#) | 2023-08-04

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[JHEP 01 \(2024\) 173](#) | [arXiv:2308.01253](#) | [DOI:10.1007/JHEP01\(2024\)173](#) | 2023-08-02

309. Search for new physics in multijet events with at least one photon and large missing transverse momentum in proton-proton collisions at 13 TeV

Hayrapetyan, Aram et al.

[JHEP 10 \(2023\) 046](#) | [arXiv:2307.16216](#) | [DOI:10.1007/JHEP10\(2023\)046](#) | 2023-07-30

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Hayrapetyan, Aram et al.

[JHEP 12 \(2023\) 068](#) | [arXiv:2307.15761](#) | [DOI:10.1007/JHEP12\(2023\)068](#) | 2023-07-28

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Tumasyan, Armen et al.

[JHEP 08 \(2024\) 148](#) | [arXiv:2307.11185](#) | [DOI:10.1007/JHEP08\(2024\)148](#) | 2023-07-20

312. Study of charm hadronization with prompt Λ_c^+ baryons in proton-proton and lead-lead collisions at $\sqrt{s_{NN}} = 5.02 \text{ TeV}$

Tumasyan, Armen et al.

[JHEP 01 \(2024\) 128](#) | [arXiv:2307.11186](#) | [DOI:10.1007/JHEP01\(2024\)128](#) | 2023-07-20

313. Search for a high-mass dimuon resonance produced in association with b quark jets at $\sqrt{s} = 13 \text{ TeV}$

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[JHEP 10 \(2023\) 043](#) | [arXiv:2307.08708](#) | [DOI:10.1007/JHEP10\(2023\)043](#) | 2023-07-14

314. Search for Z' bosons decaying to pairs of heavy Majorana neutrinos in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

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[JHEP 11 \(2023\) 181](#) | [arXiv:2307.06959](#) | [DOI:10.1007/JHEP11\(2023\)181](#) | 2023-07-12

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Atif, Mohammad et al.

[arXiv:2306.15869](#) | 2023-06-27

316. Detector R D needs for the next generation e^+e^- collider

Apresyan, A. et al.

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317. Two-particle Bose-Einstein correlations and their Lévy parameters in PbPb collisions at $\sqrt{s_{NN}}=5.02$ TeV

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[Phys.Rev.C 109 \(2024\) 024914](#) | [arXiv:2306.11574](#) | [DOI:10.1103/PhysRevC.109.024914](#) | 2023-06-20

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Tumasyan, Armen et al.

[JINST 18 \(2023\) P11017](#) | [arXiv:2306.10355](#) | [DOI:10.1088/1748-0221/18/11/P11017](#) | 2023-06-17

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[Phys.Rev.Lett. 132 \(2024\) 111901](#) | [arXiv:2306.07164](#) | [DOI:10.1103/PhysRevLett.132.111901](#) | 2023-06-12

320. Search for the lepton-flavor violating decay of the Higgs boson and additional Higgs bosons in the $e\mu$ final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[Phys.Rev.D 108 \(2023\) 072004](#) | [arXiv:2305.18106](#) | [DOI:10.1103/PhysRevD.108.072004](#) | 2023-05-29

321. Measurements of the azimuthal anisotropy of prompt and nonprompt charmonia in PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

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[JHEP 10 \(2023\) 115](#) | [arXiv:2305.16928](#) | [DOI:10.1007/JHEP10\(2023\)115](#) | 2023-05-26

322. Observation of four top quark production in proton-proton collisions at $\sqrt{s}=13$ TeV

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[Phys.Lett.B 847 \(2023\) 138290](#) | [arXiv:2305.13439](#) | [DOI:10.1016/j.physletb.2023.138290](#) | 2023-05-22

323. Search for Inelastic Dark Matter in Events with Two Displaced Muons and Missing Transverse Momentum in Proton-Proton Collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 132 \(2024\) 041802](#) | [arXiv:2305.11649](#) | [DOI:10.1103/PhysRevLett.132.041802](#) | 2023-05-19

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Tumasyan, Armen et al.

[JHEP 12 \(2023\) 189](#) | [arXiv:2305.07998](#) | [DOI:10.1007/JHEP12\(2023\)189](#) | 2023-05-13

325. Measurements of inclusive and differential cross sections for the Higgs boson production and decay to four-leptons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 08 \(2023\) 040](#) | [arXiv:2305.07532](#) | [DOI:10.1007/JHEP08\(2023\)040](#) | 2023-05-12

326. Automated Network Services for Exascale Data Movement

Balcas, Justas et al.

[EPJ Web Conf. 295 \(2024\) 01009](#) | [DOI:10.1051/epjconf/202429501009](#) | 2024

327. Observation of the rare decay of the η meson to four muons

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 131 \(2023\) 091903](#) | [arXiv:2305.04904](#) | [DOI:10.1103/PhysRevLett.131.091903](#) | 2023-05-08

328. Search for top squark pair production in a final state with at least one hadronically decaying tau lepton in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 110](#) | [arXiv:2304.07174](#) | [DOI:10.1007/JHEP07\(2023\)110](#) | 2023-04-14

329. Observation of the $\chi(3S)$ Meson and Suppression of χ States in Pb-Pb Collisions at $\sqrt{s_{NN}}=5.02\text{TeV}$

Tumasyan, Armen et al.

[Phys.Rev.Lett. 133 \(2024\) 022302](#) | [arXiv:2303.17026](#) | [DOI:10.1103/PhysRevLett.133.022302](#) | 2023-03-29

330. Probing Small Bjorken- x Nuclear Gluonic Structure via Coherent J/ψ Photoproduction in Ultraperipheral Pb-Pb Collisions at $\sqrt{s_{NN}}=5.02\text{TeV}$

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 262301](#) | [arXiv:2303.16984](#) | [DOI:10.1103/PhysRevLett.131.262301](#) | 2023-03-29

331. First measurement of the top quark pair production cross section in proton-proton collisions at $\sqrt{s} = 13.6 \text{ TeV}$

Tumasyan, Armen et al.

[JHEP 08 \(2023\) 204](#) | [arXiv:2303.10680](#) | [DOI:10.1007/JHEP08\(2023\)204](#) | 2023-03-19

332. A search for new physics in central exclusive production using the missing mass technique with the CMS detector and the CMS-TOTEM precision proton spectrometer

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 827](#) | [arXiv:2303.04596](#) | [DOI:10.1140/epjc/s10052-023-11687-5](#) | 2023-03-08

333. Evidence for Four-Top Quark Production in Proton-Proton Collisions at $\sqrt{s}=13 \text{ TeV}$

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[Phys.Lett.B 844 \(2023\) 138076](#) | [arXiv:2303.03864](#) | [DOI:10.1016/j.physletb.2023.138076](#) | 2023-03-07

334. A search for decays of the Higgs boson to invisible particles in events with a top-antitop quark pair or a vector boson in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

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[Eur.Phys.J.C 83 \(2023\) 933](#) | [arXiv:2303.01214](#) | [DOI:10.1140/epjc/s10052-023-11952-7](#) | 2023-03-02

335. Search for a vector-like quark $T \rightarrow tH$ via the diphoton decay mode of the Higgs boson in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

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[JHEP 09 \(2023\) 057](#) | [arXiv:2302.12802](#) | [DOI:10.1007/JHEP09\(2023\)057](#) | 2023-02-24

336. Measurement of the top quark mass using a profile likelihood approach with the lepton + jets final states in proton-proton collisions at $\sqrt{s}=13 \text{ TeV}$

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[Eur.Phys.J.C 83 \(2023\) 963](#) | [arXiv:2302.01967](#) | [DOI:10.1140/epjc/s10052-023-12050-4](#) | 2023-02-03

337. IRIS-HEP Strategic Plan for the Next Phase of Software Upgrades for HL-LHC Physics

Bockelman, Brian et al.

[arXiv:2302.01317](#) | 2023-02-02

338. Search for top squarks in the four-body decay mode with single lepton final states in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

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[JHEP 06 \(2023\) 060](#) | [arXiv:2301.08096](#) | [DOI:10.1007/JHEP06\(2023\)060](#) | 2023-01-19

339. First Measurement of the Forward Rapidity Gap Distribution in pPb Collisions at $\sqrt{s_{NN}}=8.16\text{TeV}$

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[Phys.Rev.D 108 \(2023\) 092004](#) | [arXiv:2301.07630](#) | [DOI:10.1103/PhysRevD.108.092004](#) | 2023-01-18

340. K_S^0 and $\Lambda(\Lambda^0)$ two-particle femtoscopic correlations in PbPb collisions at $\sqrt{s_{NN}}=5.02 \text{ TeV}$

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[Phys.Lett.B 857 \(2024\) 138936](#) | [arXiv:2301.05290](#) | [DOI:10.1016/j.physletb.2024.138936](#) | 2023-01-12

341. Measurement of the electroweak production of $W\gamma$ in association with two jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 108 \(2023\) 032017](#) | [arXiv:2212.12592](#) | [DOI:10.1103/PhysRevD.108.032017](#) | 2022-12-23

342. Search for new physics in the τ lepton plus missing transverse momentum final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[JHEP 09 \(2023\) 051](#) | [arXiv:2212.12604](#) | [DOI:10.1007/JHEP09\(2023\)051](#) | 2022-12-23

343. Measurement of the $B^0_S \rightarrow \mu^+ \mu^-$ decay properties and search for the $B^0 \rightarrow \mu^+ \mu^-$ decay in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[Phys.Lett.B 842 \(2023\) 137955](#) | [arXiv:2212.10311](#) | [DOI:10.1016/j.physletb.2023.137955](#) | 2022-12-20

344. Search for long-lived particles using out-of-time trackless jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 210](#) | [arXiv:2212.06695](#) | [DOI:10.1007/JHEP07\(2023\)210](#) | 2022-12-13

345. Measurement of the dependence of the hadron production fraction ratio f_s / f_u and f_d / f_u on B meson kinematic variables in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[Phys.Rev.Lett. 131 \(2023\) 121901](#) | [arXiv:2212.02309](#) | [DOI:10.1103/PhysRevLett.131.121901](#) | 2022-12-05

346. Measurements of azimuthal anisotropy of nonprompt D0 mesons in PbPb collisions at $\sqrt{s_{NN}}=5.02$ TeV

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[Phys.Lett.B 850 \(2024\) 138389](#) | [arXiv:2212.01636](#) | [DOI:10.1016/j.physletb.2023.138389](#) | 2022-12-03

347. Search for high-mass exclusive $\gamma\gamma \rightarrow WW$ and $\gamma\gamma \rightarrow ZZ$ production in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 229](#) | [arXiv:2211.16320](#) | [DOI:10.1007/JHEP07\(2023\)229](#) | 2022-11-29

348. Search for Higgs Boson and Observation of Z Boson through their Decay into a Charm Quark-Antiquark Pair in Boosted Topologies in Proton-Proton Collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 041801](#) | [arXiv:2211.14181](#) | [DOI:10.1103/PhysRevLett.131.041801](#) | 2022-11-25

349. Search for supersymmetry in final states with a single electron or muon using angular correlations and heavy-object identification in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 09 \(2023\) 149](#) | [arXiv:2211.08476](#) | [DOI:10.1007/JHEP09\(2023\)149](#) | 2022-11-15

350. Measurement of the differential $t\bar{t}$ production cross section as a function of the jet mass and extraction of the top quark mass in hadronic decays of boosted top quarks

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[Eur.Phys.J.C 83 \(2023\) 560](#) | [arXiv:2211.01456](#) | [DOI:10.1140/epjc/s10052-023-11587-8](#) | 2022-11-02

351. Azimuthal correlations in Z +jets events in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[Eur.Phys.J.C 83 \(2023\) 722](#) | [arXiv:2210.16139](#) | [DOI:10.1140/epjc/s10052-023-11833-z](#) | 2022-10-28

352. Application of Portable Parallelization Strategies for GPUs on track reconstruction kernels

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[JHEP 05 \(2022\) 014](#) | [arXiv:2201.04206](#) | [DOI:10.1007/JHEP05\(2022\)014](#) | 2022-01-11

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[Phys.Rev.Lett. 128 \(2022\) 252301](#) | [arXiv:2201.02659](#) | [DOI:10.1103/PhysRevLett.128.252301](#) | 2022-01-07

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[JHEP 04 \(2022\) 144](#) | [arXiv:2112.09114](#) | [DOI:10.1007/JHEP04\(2022\)144](#) | 2021-12-16

437. Evidence for WW/WZ vector boson scattering in the decay channel $\ell\nu qq$ produced in association with two jets in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 834 \(2022\) 137438](#) | [arXiv:2112.05259](#) | [DOI:10.1016/j.physletb.2022.137438](#) | 2021-12-09

438. Search for a right-handed W boson and a heavy neutrino in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 04 \(2022\) 047](#) | [arXiv:2112.03949](#) | [DOI:10.1007/JHEP04\(2022\)047](#) | 2021-12-07

439. Search for heavy resonances decaying to a pair of Lorentz-boosted Higgs bosons in final states with leptons and a bottom quark pair at $\sqrt{s}= 13$ TeV

Tumasyan, Armen et al.

[JHEP 05 \(2022\) 005](#) | [arXiv:2112.03161](#) | [DOI:10.1007/JHEP05\(2022\)005](#) | 2021-12-06

440. Measurements of the associated production of a W boson and a charm quark in proton-proton collisions at $\sqrt{s}=8$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 82 \(2022\) 1094](#) | [arXiv:2112.00895](#) | [DOI:10.1140/epjc/s10052-022-10897-7](#) | 2021-12-01

441. Measurement of $W^{\pm}\gamma$ differential cross sections in proton-proton collisions at $\sqrt{s} = 13$ TeV and effective field theory constraints

Tumasyan, Armen et al.

[Phys.Rev.D 105 \(2022\) 052003](#) | [arXiv:2111.13948](#) | [DOI:10.1103/PhysRevD.105.052003](#) | 2021-11-27

442. Search for heavy resonances decaying to ZZ or ZW and axion-like particles mediating nonresonant ZZ or ZH production at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 04 \(2022\) 087](#) | [arXiv:2111.13669](#) | [DOI:10.1007/JHEP04\(2022\)087](#) | 2021-11-26

443. Search for a heavy resonance decaying into a top quark and a W boson in the lepton+jets final state at $\sqrt{s} = 13$ TeV

Tumasyan, A. et al.

[JHEP 04 \(2022\) 048](#) | [arXiv:2111.10216](#) | [DOI:10.1007/JHEP04\(2022\)048](#) | 2021-11-19

444. Measurement and QCD analysis of double-differential inclusive jet cross sections in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 02 \(2022\) 142](#) | [arXiv:2111.10431](#) | [DOI:10.1007/JHEP02\(2022\)142](#) | 2021-11-19

445. Strategies and performance of the CMS silicon tracker alignment during LHC Run 2

Tumasyan, Armen et al.

[Nucl.Instrum.Meth.A 1037 \(2022\) 166795](#) | [arXiv:2111.08757](#) | [DOI:10.1016/j.nima.2022.166795](#) | 2021-11-16

446. Search for supersymmetry in final states with two or three soft leptons and missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 04 \(2022\) 091](#) | [arXiv:2111.06296](#) | [DOI:10.1007/JHEP04\(2022\)091](#) | 2021-11-11

447. Observation of triple J/ψ meson production in proton-proton collisions

Tumasyan, Armen et al.

[Nature Phys. 19 \(2023\) 338](#) | [arXiv:2111.05370](#) | [DOI:10.1038/s41567-022-01838-y](#) | 2021-11-09

448. Study of dijet events with large rapidity separation in proton-proton collisions at $\sqrt{s} = 2.76$ TeV

Tumasyan, Armen et al.

[JHEP 03 \(2022\) 189](#) | [arXiv:2111.04605](#) | [DOI:10.1007/JHEP03\(2022\)189](#) | 2021-11-08

449. Inclusive and differential cross section measurements of single top quark production in association with a Z boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 02 \(2022\) 107](#) | [arXiv:2111.02860](#) | [DOI:10.1007/JHEP02\(2022\)107](#) | 2021-11-04

450. A new calibration method for charm jet identification validated with proton-proton collision events at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JINST 17 \(2022\) P03014](#) | [arXiv:2111.03027](#) | [DOI:10.1088/1748-0221/17/03/P03014](#) | 2021-11-04

451. Search for Flavor-Changing Neutral Current Interactions of the Top Quark and Higgs Boson in Final States with Two Photons in Proton-Proton Collisions at $\sqrt{s} = 13$ TeV

Tumasyan, A. et al.

[Phys.Rev.Lett. 129 \(2022\) 032001](#) | [arXiv:2111.02219](#) | [DOI:10.1103/PhysRevLett.129.032001](#) | 2021-11-03

452. Search for low-mass dilepton resonances in Higgs boson decays to four-lepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, A. et al.

[Eur.Phys.J.C 82 \(2022\) 290](#) | [arXiv:2111.01299](#) | [DOI:10.1140/epjc/s10052-022-10127-0](#) | 2021-11-01

453. Search for long-lived particles produced in association with a Z boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 03 \(2022\) 160](#) | [arXiv:2110.13218](#) | [DOI:10.1007/JHEP03\(2022\)160](#) | 2021-10-25

454. Measurement of the inclusive and differential WZ production cross sections, polarization angles, and triple gauge couplings in pp collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2022\) 032](#) | [arXiv:2110.11231](#) | [DOI:10.1007/JHEP07\(2022\)032](#) | 2021-10-21

455. First Search for Exclusive Diphoton Production at High Mass with Tagged Protons in Proton-Proton Collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 129 \(2022\) 011801](#) | [arXiv:2110.05916](#) | [DOI:10.1103/PhysRevLett.129.011801](#) | 2021-10-12

456. Analysis of the CP structure of the Yukawa coupling between the Higgs boson and τ leptons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 06 \(2022\) 012](#) | [arXiv:2110.04836](#) | [DOI:10.1007/JHEP06\(2022\)012](#) | 2021-10-10

457. Search for long-lived particles decaying to leptons with large impact parameter in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 82 \(2022\) 153](#) | [arXiv:2110.04809](#) | [DOI:10.1140/epjc/s10052-022-10027-3](#) | 2021-10-10

458. Measurement of double-parton scattering in inclusive production of four jets with low transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 01 \(2022\) 177](#) | [arXiv:2109.13822](#) | [DOI:10.1007/JHEP01\(2022\)177](#) | 2021-09-28

459. Search for heavy resonances decaying to $Z(\nu\bar{\nu})V(q\bar{q}')$ in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 106 \(2022\) 012004](#) | [arXiv:2109.08268](#) | [DOI:10.1103/PhysRevD.106.012004](#) | 2021-09-17

460. Search for heavy resonances decaying to WW, WZ, or WH boson pairs in the lepton plus merged jet final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 105 \(2022\) 032008](#) | [arXiv:2109.06055](#) | [DOI:10.1103/PhysRevD.105.032008](#) | 2021-09-13

461. Study of quark and gluon jet substructure in Z+jet and dijet events from pp collisions

Tumasyan, Armen et al.

[JHEP 01 \(2022\) 188](#) | [arXiv:2109.03340](#) | [DOI:10.1007/JHEP01\(2022\)188](#) | 2021-09-07

462. Observation of B^0_s mesons and measurement of the B^0_s/B^+ yield ratio in PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 829 \(2022\) 137062](#) | [arXiv:2109.01908](#) | [DOI:10.1016/j.physletb.2022.137062](#) | 2021-09-04

463. Observation of tW production in the single-lepton channel in pp collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 11 \(2021\) 111](#) | [arXiv:2109.01706](#) | [DOI:10.1007/JHEP11\(2021\)111](#) | 2021-09-03

464. Measurement of the top quark mass using events with a single reconstructed top quark in pp collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 12 \(2021\) 161](#) | [arXiv:2108.10407](#) | [DOI:10.1007/JHEP12\(2021\)161](#) | 2021-08-23

465. Measurement of differential $t\bar{t}$ production cross sections in the full kinematic range using lepton+jets events from proton-proton collisions at $\sqrt{s} = 13$ TeV

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[Phys.Rev.D 104 \(2021\) 092013](#) | [arXiv:2108.02803](#) | [DOI:10.1103/PhysRevD.104.092013](#) | 2021-08-05

466. Probing effective field theory operators in the associated production of top quarks with a Z boson in multilepton final states at $\sqrt{s} = 13$ TeV

Lee, Kyeongpil et al.

[JHEP 12 \(2021\) 083](#) | [arXiv:2107.13896](#) | [DOI:10.1007/JHEP12\(2021\)083](#) | 2021-07-29

467. Search for new particles in events with energetic jets and large missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 11 \(2021\) 153](#) | [arXiv:2107.13021](#) | [DOI:10.1007/JHEP11\(2021\)153](#) | 2021-07-27

468. Search for chargino-neutralino production in events with Higgs and W bosons using 137 fb^{-1} of proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 10 \(2021\) 045](#) | [arXiv:2107.12553](#) | [DOI:10.1007/JHEP10\(2021\)045](#) | 2021-07-27

469. Measurement of the inclusive and differential Higgs boson production cross sections in the decay mode to a pair of τ leptons in pp collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 128 \(2022\) 081805](#) | [arXiv:2107.11486](#) | [DOI:10.1103/PhysRevLett.128.081805](#) | 2021-07-23

470. Combined searches for the production of supersymmetric top quark partners in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 81 \(2021\) 970](#) | [arXiv:2107.10892](#) | [DOI:10.1140/epjc/s10052-021-09721-5](#) | 2021-07-22

471. Search for Long-Lived Particles Decaying in the CMS End Cap Muon Detectors in Proton-Proton Collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 127 \(2021\) 261804](#) | [arXiv:2107.04838](#) | [DOI:10.1103/PhysRevLett.127.261804](#) | 2021-07-10

472. Measurement of prompt open-charm production cross sections in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 11 \(2021\) 225](#) | [arXiv:2107.01476](#) | [DOI:10.1007/JHEP11\(2021\)225](#) | 2021-07-03

473. Measurement of the inclusive and differential $t\bar{t}\gamma$ cross sections in the single-lepton channel and EFT interpretation at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 12 \(2021\) 180](#) | [arXiv:2107.01508](#) | [DOI:10.1007/JHEP12\(2021\)180](#) | 2021-07-03

474. Measurements of the electroweak diboson production cross sections in proton-proton collisions at $\sqrt{s} = 5.02$ TeV using leptonic decays

Tumasyan, Armen et al.

[Phys.Rev.Lett. 127 \(2021\) 191801](#) | [arXiv:2107.01137](#) | [DOI:10.1103/PhysRevLett.127.191801](#) | 2021-07-02

475. Search for electroweak production of charginos and neutralinos in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 04 \(2022\) 147](#) | [arXiv:2106.14246](#) | [DOI:10.1007/JHEP04\(2022\)147](#) | 2021-06-27

476. Fragmentation of jets containing a prompt J/ψ meson in PbPb and pp collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 825 \(2022\) 136842](#) | [arXiv:2106.13235](#) | [DOI:10.1016/j.physletb.2021.136842](#) | 2021-06-23

477. Measurement of the electroweak production of $Z\gamma$ and two jets in proton-proton collisions at $\sqrt{s} = 13$ TeV and constraints on anomalous quartic gauge couplings

Tumasyan, Armen et al.

[Phys.Rev.D 104 \(2021\) 072001](#) | [arXiv:2106.11082](#) | [DOI:10.1103/PhysRevD.104.072001](#) | 2021-06-21

478. Search for $W\gamma$ resonances in proton-proton collisions at $\sqrt{s} = 13$ TeV using hadronic decays of Lorentz-boosted W bosons

Tumasyan, Armen et al.

[Phys.Lett.B 826 \(2022\) 136888](#) | [arXiv:2106.10509](#) | [DOI:10.1016/j.physletb.2022.136888](#) | 2021-06-19

479. Search for a heavy Higgs boson decaying into two lighter Higgs bosons in the $\tau\tau b\bar{b}$ final state at 13 TeV

Tumasyan, Armen et al.

[JHEP 11 \(2021\) 057](#) | [arXiv:2106.10361](#) | [DOI:10.1007/JHEP11\(2021\)057](#) | 2021-06-18

480. Study of Z boson plus jets events using variables sensitive to double-parton scattering in pp collisions at 13 TeV

Tumasyan, Armen et al.

[JHEP 10 \(2021\) 176](#) | [arXiv:2105.14511](#) | [DOI:10.1007/JHEP10\(2021\)176](#) | 2021-05-30

481. Measurements of the $pp \rightarrow W^{\pm}\gamma\gamma$ and $pp \rightarrow Z\gamma\gamma$ cross sections at $\sqrt{s} = 13$ TeV and limits on anomalous quartic gauge couplings

Tumasyan, Armen et al.

[JHEP 10 \(2021\) 174](#) | [arXiv:2105.12780](#) | [DOI:10.1007/JHEP10\(2021\)174](#) | 2021-05-26

482. Search for strongly interacting massive particles generating trackless jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 82 \(2022\) 213](#) | [arXiv:2105.09178](#) | [DOI:10.1140/epjc/s10052-022-10095-5](#) | 2021-05-19

483. Search for lepton-flavor violating decays of the Higgs boson in the $\mu\tau$ and $e\tau$ final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 104 \(2021\) 032013](#) | [arXiv:2105.03007](#) | [DOI:10.1103/PhysRevD.104.032013](#) | 2021-05-06

484. Search for long-lived particles decaying to jets with displaced vertices in proton-proton collisions at $\sqrt{s}= 13$ TeV

Sirunyan, A.M. et al.

[Phys.Rev.D 104 \(2021\) 052011](#) | [arXiv:2104.13474](#) | [DOI:10.1103/PhysRevD.104.052011](#) | 2021-04-27

485. Search for a heavy resonance decaying to a top quark and a W boson at $\sqrt{s} = 13$ TeV in the fully hadronic final state

Sirunyan, Albert M et al.

[JHEP 12 \(2021\) 106](#) | [arXiv:2104.12853](#) | [DOI:10.1007/JHEP12\(2021\)106](#) | 2021-04-26

486. Constraints on anomalous Higgs boson couplings to vector bosons and fermions in its production and decay using the four-lepton final state

Sirunyan, Albert M et al.

[Phys.Rev.D 104 \(2021\) 052004](#) | [arXiv:2104.12152](#) | [DOI:10.1103/PhysRevD.104.052004](#) | 2021-04-25

487. Search for W' bosons decaying to a top and a bottom quark at $s=13\text{TeV}$ in the hadronic final state

Sirunyan, Albert M et al.

[Phys.Lett.B 820 \(2021\) 136535](#) | [arXiv:2104.04831](#) | [DOI:10.1016/j.physletb.2021.136535](#) | 2021-04-10

488. Search for charged Higgs bosons produced in vector boson fusion processes and decaying into vector boson pairs in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 723](#) | [arXiv:2104.04762](#) | [DOI:10.1140/epjc/s10052-021-09472-3](#) | 2021-04-10

489. Precision luminosity measurement in proton-proton collisions at $\sqrt{s} = 13$ TeV in 2015 and 2016 at CMS

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 800](#) | [arXiv:2104.01927](#) | [DOI:10.1140/epjc/s10052-021-09538-2](#) | 2021-04-05

490. Constraints on the Initial State of Pb-Pb Collisions via Measurements of Z-Boson Yields and Azimuthal Anisotropy at $\sqrt{s_{NN}}=5.02$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 127 \(2021\) 102002](#) | [arXiv:2103.14089](#) | [DOI:10.1103/PhysRevLett.127.102002](#) | 2021-03-25

491. Measurements of Higgs boson production cross sections and couplings in the diphoton decay channel at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 07 \(2021\) 027](#) | [arXiv:2103.06956](#) | [DOI:10.1007/JHEP07\(2021\)027](#) | 2021-03-11

492. Measurements of production cross sections of the Higgs boson in the four-lepton final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 488](#) | [arXiv:2103.04956](#) | [DOI:10.1140/epjc/s10052-021-09200-x](#) | 2021-03-08

493. Using Z Boson Events to Study Parton-Medium Interactions in Pb-Pb Collisions

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 128 \(2022\) 122301](#) | [arXiv:2103.04377](#) | [DOI:10.1103/PhysRevLett.128.122301](#) | 2021-03-07

494. Search for resonant and nonresonant new phenomena in high-mass dilepton final states at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 07 \(2021\) 208](#) | [arXiv:2103.02708](#) | [DOI:10.1007/JHEP07\(2021\)208](#) | 2021-03-03

495. Search for top squark production in fully-hadronic final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 104 \(2021\) 052001](#) | [arXiv:2103.01290](#) | [DOI:10.1103/PhysRevD.104.052001](#) | 2021-03-01

496. Study of Drell-Yan dimuon production in proton-lead collisions at $\sqrt{s_{NN}} = 8.16$ TeV

Sirunyan, Albert M et al.

[JHEP 05 \(2021\) 182](#) | [arXiv:2102.13648](#) | [DOI:10.1007/JHEP05\(2021\)182](#) | 2021-02-26

497. Evidence for X(3872) in Pb-Pb Collisions and Studies of its Prompt Production at $\sqrt{s_{NN}}=5.02$ TeV

Sirunyan, Albert M. et al.

[Phys.Rev.Lett. 128 \(2022\) 032001](#) | [arXiv:2102.13048](#) | [DOI:10.1103/PhysRevLett.128.032001](#) | 2021-02-25

498. First measurement of large area jet transverse momentum spectra in heavy-ion collisions

Sirunyan, Albert M et al.

[JHEP 05 \(2021\) 284](#) | [arXiv:2102.13080](#) | [DOI:10.1007/JHEP05\(2021\)284](#) | 2021-02-25

499. Measurements of angular distance and momentum ratio distributions in three-jet and Z + two-jet final states in pp collisions

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 852](#) | [arXiv:2102.08816](#) | [DOI:10.1140/epjc/s10052-021-09570-2](#) | 2021-02-17

500. Search for a heavy vector resonance decaying to a Z^0 boson and a Higgs boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 688](#) | [arXiv:2102.08198](#) | [DOI:10.1140/epjc/s10052-021-09348-6](#) | 2021-02-16

501. Hard color-singlet exchange in dijet events in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 104 \(2021\) 032009](#) | [arXiv:2102.06945](#) | [DOI:10.1103/PhysRevD.104.032009](#) | 2021-02-13

502. Search for top squarks in final states with two top quarks and several light-flavor jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 104 \(2021\) 032006](#) | [arXiv:2102.06976](#) | [DOI:10.1103/PhysRevD.104.032006](#) | 2021-02-13

503. Performance of the CMS muon trigger system in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JINST 16 \(2021\) P07001](#) | [arXiv:2102.04790](#) | [DOI:10.1088/1748-0221/16/07/P07001](#) | 2021-02-09

504. Observation of a New Excited Beauty Strange Baryon Decaying to $\Xi^0 \Lambda^0 \pi^0$

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 126 \(2021\) 252003](#) | [arXiv:2102.04524](#) | [DOI:10.1103/PhysRevLett.126.252003](#) | 2021-02-08

505. Measurements of the differential cross sections of the production of Z + jets and γ + jets and of Z boson emission collinear with a jet in pp collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 05 \(2021\) 285](#) | [arXiv:2102.02238](#) | [DOI:10.1007/JHEP05\(2021\)285](#) | 2021-02-03

506. Measurement of the $W\gamma$ Production Cross Section in Proton-Proton Collisions at $\sqrt{s}=13$ TeV and Constraints on Effective Field Theory Coefficients

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 126 \(2021\) 252002](#) | [arXiv:2102.02283](#) | [DOI:10.1103/PhysRevLett.126.252002](#) | 2021-02-03

507. In-medium modification of dijets in PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Sirunyan, Albert M et al.

[JHEP 05 \(2021\) 116](#) | [arXiv:2101.04720](#) | [DOI:10.1007/JHEP05\(2021\)116](#) | 2021-01-12

508. First measurement of the cross section for top quark pair production with additional charm jets using dileptonic final states in pp collisions at $s=13\text{TeV}$

Sirunyan, Albert M et al.

[Phys.Lett.B 820 \(2021\) 136565](#) | [arXiv:2012.09225](#) | [DOI:10.1016/j.physletb.2021.136565](#) | 2020-12-16

509. Measurement of the Z boson differential production cross section using its invisible decay mode ($Z\nu\bar{\nu}$) in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 05 \(2021\) 205](#) | [arXiv:2012.09254](#) | [DOI:10.1007/JHEP05\(2021\)205](#) | 2020-12-16

510. Search for supersymmetry in final states with two oppositely charged same-flavor leptons and missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 04 \(2021\) 123](#) | [arXiv:2012.08600](#) | [DOI:10.1007/JHEP04\(2021\)123](#) | 2020-12-15

511. Electron and photon reconstruction and identification with the CMS experiment at the CERN LHC

Sirunyan, Albert M et al.

[JINST 16 \(2021\) P05014](#) | [arXiv:2012.06888](#) | [DOI:10.1088/1748-0221/16/05/P05014](#) | 2020-12-12

512. Measurement of differential cross sections for Z bosons produced in association with charm jets in pp collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 04 \(2021\) 109](#) | [arXiv:2012.04119](#) | [DOI:10.1007/JHEP04\(2021\)109](#) | 2020-12-07

513. Search for singly and pair-produced leptoquarks coupling to third-generation fermions in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Sirunyan, Albert M et al.

[Phys.Lett.B 819 \(2021\) 136446](#) | [arXiv:2012.04178](#) | [DOI:10.1016/j.physletb.2021.136446](#) | 2020-12-07

514. Search for new physics in top quark production with additional leptons in proton-proton collisions at $\sqrt{s} = 13$ TeV using effective field theory

Sirunyan, Albert M et al.

[JHEP 03 \(2021\) 095](#) | [arXiv:2012.04120](#) | [DOI:10.1007/JHEP03\(2021\)095](#) | 2020-12-07

515. Search for long-lived particles using displaced jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 104 \(2021\) 012015](#) | [arXiv:2012.01581](#) | [DOI:10.1103/PhysRevD.104.012015](#) | 2020-12-02

516. Search for nonresonant Higgs boson pair production in final states with two bottom quarks and two photons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 03 \(2021\) 257](#) | [arXiv:2011.12373](#) | [DOI:10.1007/JHEP03\(2021\)257](#) | 2020-11-24

517. Search for the rare decay of the W boson into a pion and a photon in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 819 \(2021\) 136409](#) | [arXiv:2011.06028](#) | [DOI:10.1016/j.physletb.2021.136409](#) | 2020-11-11

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Sirunyan, Albert M et al.

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Sirunyan, Albert M. et al.

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Sirunyan, Albert M et al.

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Sirunyan, Albert M et al.

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[Phys.Rev.Lett. 121 \(2018\) 092002](#) | [arXiv:1805.11192](#) | [DOI:10.1103/PhysRevLett.121.092002](#) | 2018-05-28

779. Search for an exotic decay of the Higgs boson to a pair of light pseudoscalars in the final state with two b quarks and two τ leptons in proton-proton collisions at $\sqrt{s} = 13$ TeV

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Sirunyan, Albert M et al.

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[Phys.Lett.B 790 \(2019\) 270](#) | [arXiv:1805.09215](#) | [DOI:10.1016/j.physletb.2019.01.006](#) | 2018-05-23

782. Measurement of the production cross section for single top quarks in association with W bosons in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

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783. Search for top squarks decaying via four-body or chargino-mediated modes in single-lepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, A.M. et al.

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784. Search for black holes and sphalerons in high-multiplicity final states in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

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Sirunyan, Albert M et al.

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786. Search for an exotic decay of the Higgs boson to a pair of light pseudoscalars in the final state of two muons and two τ leptons in proton-proton collisions at $\sqrt{s}=13$ TeV

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[JHEP 11 \(2018\) 018](#) | [arXiv:1805.04865](#) | [DOI:10.1007/JHEP11\(2018\)018](#) | 2018-05-13

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Sirunyan, Albert M et al.

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[Phys.Lett.B 790 \(2019\) 509](#) | [arXiv:1805.02248](#) | [DOI:10.1016/j.physletb.2019.01.058](#) | 2018-05-06

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[Phys.Rev.Lett. 121 \(2018\) 082301](#) | [arXiv:1804.09767](#) | [DOI:10.1103/PhysRevLett.121.082301](#) | 2018-04-25

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[JHEP 08 \(2018\) 016](#) | [arXiv:1804.07321](#) | [DOI:10.1007/JHEP08\(2018\)016](#) | 2018-04-19

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[Eur.Phys.J.C 78 \(2018\) 965](#) | [arXiv:1804.05252](#) | [DOI:10.1140/epjc/s10052-018-6373-0](#) | 2018-04-14

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[JINST 13 \(2018\) P06015](#) | [arXiv:1804.04528](#) | [DOI:10.1088/1748-0221/13/06/P06015](#) | 2018-04-12

795. Search for ttH production in the $H \rightarrow b\bar{b}$ decay channel with leptonic tt decays in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

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[Phys.Rev.Lett. 120 \(2018\) 231801](#) | [arXiv:1804.02610](#) | [DOI:10.1103/PhysRevLett.120.231801](#) | 2018-04-07

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Sirunyan, Albert M et al.

[JHEP 05 \(2018\) 148](#) | [arXiv:1803.11116](#) | [DOI:10.1007/JHEP05\(2018\)148](#) | 2018-03-29

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802. Search for a heavy resonance decaying into a Z boson and a Z or W boson in $2\ell 2q$ final states at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 09 \(2018\) 101](#) | [arXiv:1803.10093](#) | [DOI:10.1007/JHEP09\(2018\)101](#) | 2018-03-27

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Sirunyan, Albert M et al.

[Phys.Rev.D 97 \(2018\) 112003](#) | [arXiv:1803.08856](#) | [DOI:10.1103/PhysRevD.97.112003](#) | 2018-03-23

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Sirunyan, Albert M et al.

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[JHEP 06 \(2018\) 101](#) | [arXiv:1803.06986](#) | [DOI:10.1007/JHEP06\(2018\)101](#) | 2018-03-19

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[JHEP 09 \(2018\) 007](#) | [arXiv:1803.06553](#) | [DOI:10.1007/JHEP09\(2018\)007](#) | 2018-03-17

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[JHEP 06 \(2018\) 120](#) | [arXiv:1803.06292](#) | [DOI:10.1007/JHEP06\(2018\)120](#) | 2018-03-16

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Sirunyan, Albert M et al.

[JHEP 07 \(2018\) 153](#) | [arXiv:1803.04496](#) | [DOI:10.1007/JHEP07\(2018\)153](#) | 2018-03-12

810. Measurements of differential cross sections of top quark pair production as a function of kinematic event variables in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[JHEP 06 \(2018\) 002](#) | [arXiv:1803.03991](#) | [DOI:10.1007/JHEP06\(2018\)002](#) | 2018-03-11

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Sirunyan, Albert M et al.

[JHEP 07 \(2018\) 075](#) | [arXiv:1803.03838](#) | [DOI:10.1007/JHEP07\(2018\)075](#) | 2018-03-10

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[Phys.Rev.Lett. 120 \(2018\) 221801](#) | [arXiv:1802.02965](#) | [DOI:10.1103/PhysRevLett.120.221801](#) | 2018-02-08

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[JHEP 05 \(2018\) 025](#) | [arXiv:1802.02110](#) | [DOI:10.1007/JHEP05\(2018\)025](#) | 2018-02-06

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[JHEP 03 \(2018\) 160](#) | [arXiv:1801.03957](#) | [DOI:10.1007/JHEP03\(2018\)160](#) | 2018-01-11

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[Eur.Phys.J.C 78 \(2018\) 708](#) | [arXiv:1801.03535](#) | [DOI:10.1140/epjc/s10052-018-6146-9](#) | 2018-01-10

827. Search for new physics in events with two soft oppositely charged leptons and missing transverse momentum in proton-proton collisions at $\sqrt{s}=13$ TeV

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[JHEP 05 \(2018\) 127](#) | [arXiv:1801.00359](#) | [DOI:10.1007/JHEP05\(2018\)127](#) | 2017-12-31

829. Electroweak production of two jets in association with a Z boson in proton-proton collisions at $\sqrt{s}=13$ TeV

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[Eur.Phys.J.C 78 \(2018\) 589](#) | [arXiv:1712.09814](#) | [DOI:10.1140/epjc/s10052-018-6049-9](#) | 2017-12-28

830. Search for R-parity violating supersymmetry in pp collisions at $\sqrt{s}=13$ TeV using b jets in a final state with a single lepton, many jets, and high sum of large-radius jet masses

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[Phys.Lett.B 783 \(2018\) 114](#) | [arXiv:1712.08920](#) | [DOI:10.1016/j.physletb.2018.06.028](#) | 2017-12-24

831. Measurement of prompt and nonprompt charmonium suppression in PbPb collisions at 5.02 TeV

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[Eur.Phys.J.C 78 \(2018\) 509](#) | [arXiv:1712.08959](#) | [DOI:10.1140/epjc/s10052-018-5950-6](#) | 2017-12-24

832. Search for Physics Beyond the Standard Model in Events with High-Momentum Higgs Bosons and Missing Transverse Momentum in Proton-Proton Collisions at 13 TeV

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[Phys.Rev.Lett. 120 \(2018\) 241801](#) | [arXiv:1712.08501](#) | [DOI:10.1103/PhysRevLett.120.241801](#) | 2017-12-22

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[Phys.Rev.C 97 \(2018\) 064912](#) | [arXiv:1712.07198](#) | [DOI:10.1103/PhysRevC.97.064912](#) | 2017-12-19

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843. Search for the flavor-changing neutral current interactions of the top quark and the Higgs boson which decays into a pair of b quarks at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

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844. Search for pair production of excited top quarks in the lepton + jets final state

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[JHEP 07 \(2018\) 032](#) | [arXiv:1711.04299](#) | [DOI:10.1007/JHEP07\(2018\)032](#) | 2017-11-12

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853. Measurement of the inclusive tt cross section in pp collisions at $\sqrt{s}=5.02$ TeV using final states with at least one charged lepton

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Khachatryan, Vardan et al.

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Khachatryan, Vardan et al.

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[Phys.Rev.C 96 \(2017\) 014915](#) | [arXiv:1604.05347](#) | [DOI:10.1103/PhysRevC.96.014915](#) | 2016-04-18

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Khachatryan, Vardan et al.

[Phys.Lett.B 761 \(2016\) 31](#) | [arXiv:1603.02913](#) | [DOI:10.1016/j.physletb.2016.07.065](#) | 2016-03-09

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Khachatryan, Vardan et al.

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Khachatryan, Vardan et al.

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Khachatryan, Vardan et al.

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Khachatryan, Vardan et al.

[Eur.Phys.J.C 76 \(2016\) 128](#) | [arXiv:1509.06076](#) | [DOI:10.1140/epjc/s10052-016-3956-5](#) | 2015-09-20

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Khachatryan, Vardan et al.

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Khachatryan, Vardan et al.

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[Phys.Rev.Lett. 116 \(2016\) 032301](#) | [arXiv:1508.06678](#) | [DOI:10.1103/PhysRevLett.116.032301](#) | 2015-08-26

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[Phys.Lett.B 755 \(2016\) 196](#) | [arXiv:1508.04308](#) | [DOI:10.1016/j.physletb.2016.02.002](#) | 2015-08-18

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[Phys.Lett.B 752 \(2016\) 221](#) | [arXiv:1508.01437](#) | [DOI:10.1016/j.physletb.2015.11.042](#) | 2015-08-06

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[Phys.Lett.B 753 \(2016\) 424](#) | [arXiv:1507.08126](#) | [DOI:10.1016/j.physletb.2015.12.020](#) | 2015-07-29

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[Phys.Lett.B 757 \(2016\) 97](#) | [arXiv:1507.07527](#) | [DOI:10.1016/j.physletb.2016.03.046](#) | 2015-07-27

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Khachatryan, Vardan et al.

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Khachatryan, Vardan et al.

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[Phys.Rev.Lett. 114 \(2015\) 191802](#) | [arXiv:1502.04155](#) | [DOI:10.1103/PhysRevLett.114.191802](#) | 2015-02-13

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1172. Measurement of the ratio $B(B^0_s \rightarrow J/\psi f_0(980)) / B(B^0_s \rightarrow J/\psi \phi(1020))$ in pp collisions at $\sqrt{s} = 7$ TeV

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[Eur.Phys.J.C 75 \(2015\) 147](#) | [arXiv:1412.1115](#) | [DOI:10.1140/epjc/s10052-015-3364-2](#) | 2014-12-02

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Khachatryan, Vardan et al.

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[Phys.Lett.B 746 \(2015\) 79](#) | [arXiv:1411.2646](#) | [DOI:10.1016/j.physletb.2015.04.042](#) | 2014-11-10

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Khachatryan, Vardan et al.

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[Phys.Lett.B 739 \(2014\) 23](#) | [arXiv:1407.6643](#) | [DOI:10.1016/j.physletb.2014.10.032](#) | 2014-07-24

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Chatrchyan, Serguei et al.

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Chatrchyan, Serguei et al.

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Chatrchyan, Serguei et al.

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Chatrchyan, Serguei et al.

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Chatrchyan, Serguei et al.

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